Experiment No:3

Date of Performance :

Date of Submission :

AIM:-To plot the wave form for Binary Amplitude Shift Keying (BASK) signal using MATLAB for a stream of bits.

THEORY:-

Amplitude Shift Keying (ASK) is the digital modulation technique. In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes. In ASK, the amplitude of the carrier assumes one of the two amplitudes dependent on the logic states of the input bit stream. This modulated signal can be expressed as:

$$x_c(t) = \begin{cases} 0 & \text{symbol "0"} \\ A\cos\omega_c t & \text{symbol "1"} \end{cases}$$

Amplitude shift keying (ASK) in the context of digital signal communications is a modulation process, which imparts to a sinusoid two or more discrete amplitude levels. These are related to the number of levels adopted by the digital message. For a binary message sequence there are two levels, one of which is typically zero. Thus the modulated waveform consists of bursts of a sinusoid. Figure 1 illustrates a binary ASK signal (lower), together with the binary sequencewhich initiated it (upper). Neither signal has been band limited.

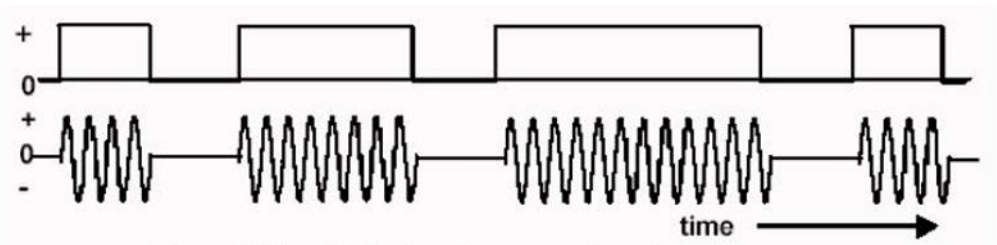


Fig:anASKsignal(below)andthemessage(above)

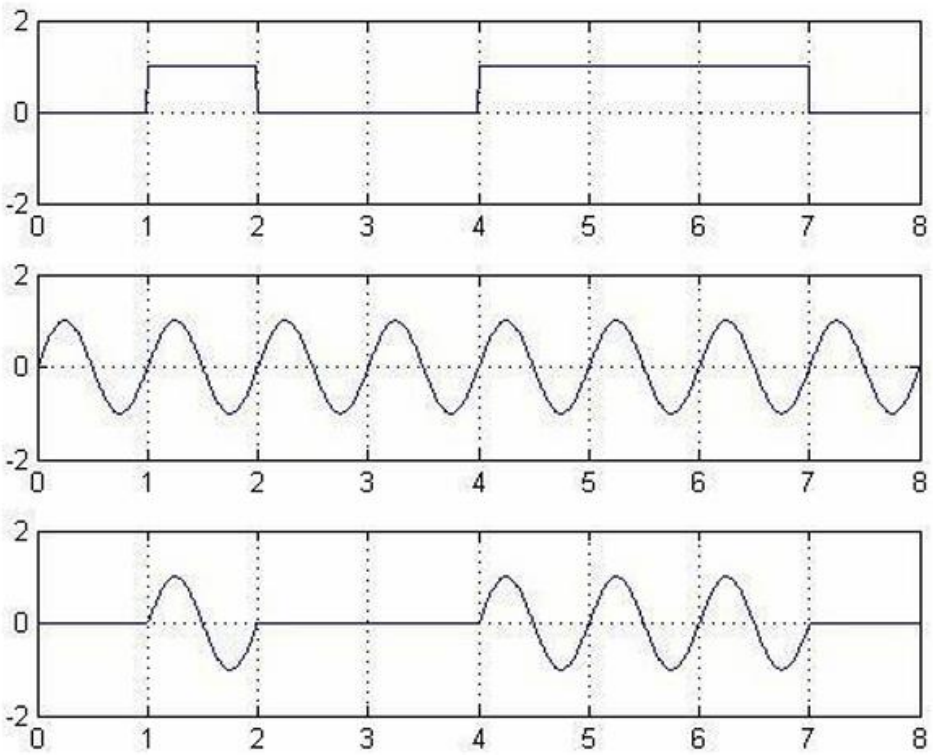
There are sharp discontinuities shown at the transition points. These result in the signal having an unnecessarily wide bandwidth. Band limiting is generally introduced before transmission, in which case these discontinuities would be 'rounded off'. The band limiting may be applied to the digital message, or the modulated signal itself. The data rate is often made a sub-multiple of the carrier frequency. This has been done in the waveform of Fig.

MATLAB PROGRAM:-

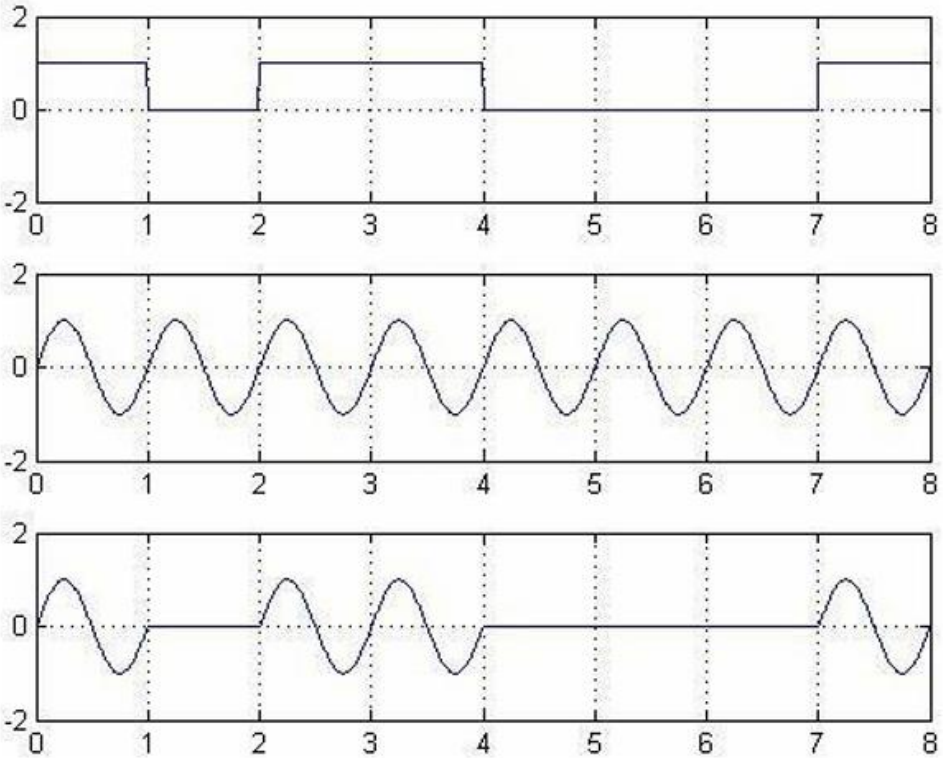
```
clear;
clc;
b=input('EntertheBitstream\n');%b=[0101110]; n =
length(b);
t = 0:.01:n;
x=1:1:(n+1)*100;
fori=1:n
   for j = i:.1:i+1
      bw(x(i*100:(i+1)*100))=b(i);
   end
end
bw=bw(100:end);
sint = sin(2*pi*t);
st = bw.*sint;
subplot(3,1,1)
plot(t,bw)
gridon;axis([0n-2+2])
subplot(3,1,2)
plot(t,sint)
gridon;axis([0n-2+2])
subplot(3,1,3)
plot(t,st)
gridon;axis([0n-2+2])
```

OBSERVATION: -

Output wave form for the bit stream [0 1 0 0 1 1 10]



Output wave form for the bit stream [1 0 1 1 0 0 01]



Result: The BASK signal was successfully generated and plotted in MATLAB for the given bit stream. The waveform showed two distinct amplitude levels corresponding to binary 1 and 0, while frequency and phase remained constant. The plotted signal verified correct Binary Amplitude Shift Keying modulation behavior.

Signature and Remark:

R1 (4 MARKS)	R2 (4 MARKS)	R3 (4 MARKS)	R4 (3 MARKS)	TOTAL MARKS (15 MARKS)	SIGNATURE